

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457305.8652 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.05
Transit depth (Rp/Rs)^2	2.05 +/- 1.44 [%]
Orbital Inclination (inc)	85.21 +/- 1.72
Airmass coefficient 1 (a1)	0.172 +/- 0.06
Airmass coefficient 2 (a2)	1.66 +/- 0.44
Scatter in the residuals of the lightcurve fit is	34.36 %
Best Comparison Star	#2 - [292, 346]
Optimal Aperture	1.7
Optimal Annulus	6.75
Transit Duration (day)	0.045 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.05 vs 0.131 (deviation 0.24σ)
Duration fit vs published	0.045 d vs 0.1075 d (deviation 1.6σ)
Mid-time O–C	3.0 min
Depth SNR	2.9
Residual scatter	34.36 %

Light curve

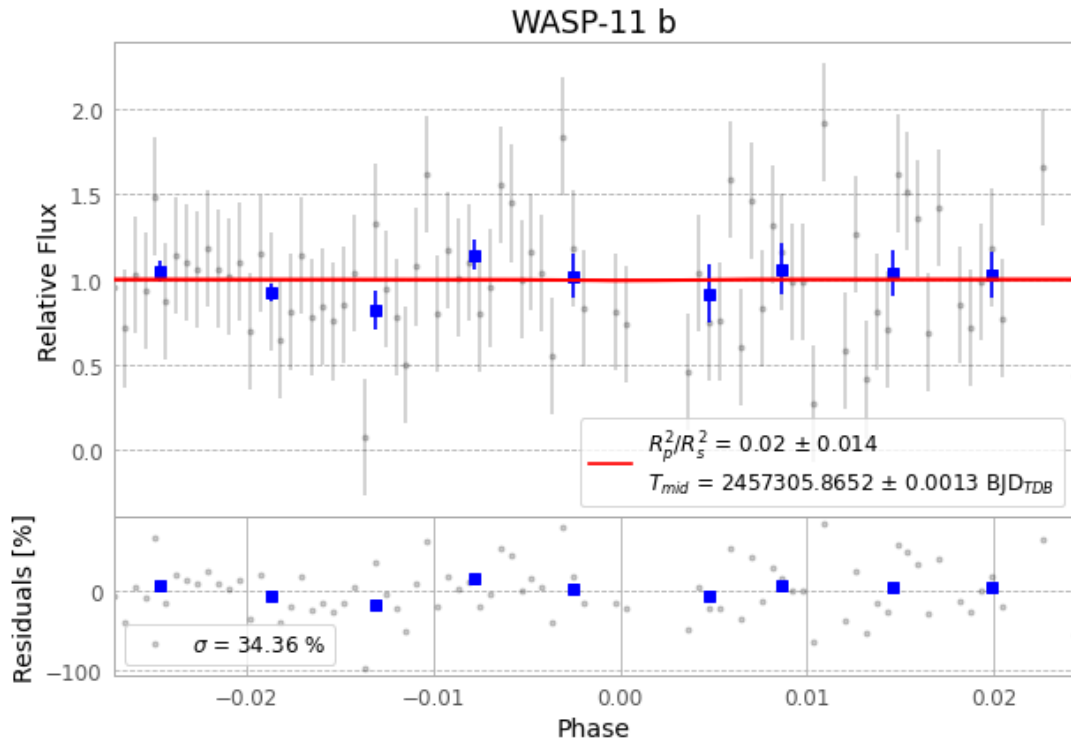


Figure 1 — FinalLightCurve_WASP-11 b_2015-10-09.png (SHA-256 e831905160a9bbda...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [632, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 321f9732630bf1f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), HATP-10151010061212.FITS → HATP-10151010105112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-151010.log (1766d56efdc3...), FinalLightCurve_WASP-11 b_2015-10-09.png (e831905160a9...), FinalLightCurve_WASP-11 b_2015-10-09.csv (989b547465d6...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 18:06Z.